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BOTTOM-UP DIRECTIONAL ATOMIC LAYER DEPOSITION (ALD)

Abstract

A method of film deposition is provided. The method includes providing a wafer including a patterned structure having a top, a bottom and a sidewall. A film can be formed on the wafer by a cyclical deposition process including a cycle of contacting the wafer with a first reactant comprising a silicon precursor to form an intermediate layer over the patterned structure of the wafer, generating a plasma comprising H_{sup.}+ ions to modify the intermediate layer by delivering the H_{sup.}+ ions anisotropically towards the top and the bottom of the patterned structure, and contacting the wafer with a second reactant to form a material layer. The silicon precursor includes a silicon-nitrogen bond and includes no halogen. The second reactant includes at least one selected from the group consisting of a nitrogen precursor, an oxygen precursor and a carbon precursor.

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Background/Summary

FIELD OF THE INVENTION

[0001] This disclosure relates generally to methods of microfabrication and more specifically to film deposition.

BACKGROUND

[0002] In the manufacture of a semiconductor device (especially on the microscopic scale), various fabrication processes are executed such as film-forming depositions, etch mask creation, patterning, material etching and removal, and doping treatments. These processes are performed repeatedly to form desired semiconductor device elements on a substrate. Historically, with microfabrication, transistors have been created in one plane, with wiring/metallization formed above the active device plane, and have thus been characterized as two-dimensional (2D) circuits or 2D fabrication. Scaling efforts have greatly increased the number of transistors per unit area and yet are running into greater challenges as miniaturization continues with three-dimensional (3D) semiconductor structure and topography.

SUMMARY

[0003] The present disclosure relates to a method of film deposition and an apparatus of executing the same.

[0004] According to a first aspect of the disclosure, a method of film deposition is provided. The method includes providing a wafer including a patterned structure having a top, a bottom and a sidewall. A film is formed on the wafer by a cyclical deposition process including a cycle of contacting the wafer with a first reactant including a silicon precursor to form an intermediate layer over the patterned structure of the wafer, generating a plasma including $H_{sup.+}$ ions to modify the intermediate layer by delivering the $H_{sup.+}$ ions anisotropically towards the top and the bottom of the patterned structure, and contacting the wafer with a second reactant to form a material layer. The silicon precursor includes a silicon-nitrogen bond and includes no halogen. The second reactant includes at least one selected from the group consisting of a nitrogen precursor, an oxygen precursor and a carbon precursor.

[0005] In some embodiments, the silicon precursor consists of silicon (Si), nitrogen (N), hydrogen (H) and carbon (C) or consists of Si, N and H.

[0006] In some embodiments, the silicon precursor includes at least one of trisilylamine (TSA) or a molecule conforming to a formula of $Si(NR_{sup.1}R_{sup.2})_{sub.n}(R_{sup.3})_{sub.4-n}$, where n is 1, 2, 3 or 4, and $R_{sup.1}$, $R_{sup.2}$, and $R_{sup.3}$ are each independently a hydrogen atom or an alkyl group having 1-18 carbon atoms.

[0007] In some embodiments, the silicon precursor includes at least one selected from the group consisting of TSA, bis(tertiary-butylamino) silane (BTBAS), bis(diethylamino) silane (BDEAS) and bis(dimethylamino)dimethylsilane (BDMADMS).

[0008] In some embodiments, the plasma is generated from a gas including at least one selected from the group consisting of $H_{sub.2}$ and a hydrocarbon.

[0009] In some embodiments, the gas includes at least one selected from the group consisting of $H_{sub.2}$, benzene, a benzene derivative having 7 to 10 carbon atoms, and an alkane having 1 to 18 carbon atoms.

[0010] In some embodiments, the gas includes $H_{sub.2}$, $CH_{sub.4}$ or a combination thereof.

[0011] In some embodiments, the plasma further includes $H_{sub.2}^{sup.+}$ and $H_{sub.3}^{sup.+}$ ions.

[0012] In some embodiments, the plasma is generated at a frequency of 100 MHz to 10 GHz.

[0013] In some embodiments, the $H_{sup.+}$ ions are substantially unidirectional so that the $H_{sup.+}$ ions are delivered substantially towards the top and the bottom of the patterned structure, relative to the sidewall of the patterned structure.

[0014] In some embodiments, the H.sup.+ ions have a substantially monotonic energy distribution that is configured to promote film deposition at the top and the bottom of the patterned structure.

[0015] In some embodiments, the cyclical deposition process includes repeating the cycle for at least one more time.

[0016] In some embodiments, the film is thicker at the top and the bottom of the patterned structure than at the sidewall of the patterned structure.

[0017] In some embodiments, the plasma is generated before or while the wafer is contacted with the second reactant.

[0018] In some embodiments, the cyclical deposition process is executed at a temperature of no higher than 500° C.

[0019] In some embodiments, the cyclical deposition process includes atomic layer deposition.

[0020] In some embodiments, —NH.sub.2 surface groups are formed on the patterned structure of the wafer.

[0021] In some embodiments, the material layer includes one selected from the group consisting of silicon nitride, silicon oxide, silicon oxynitride and silicon carbide.

[0022] In some embodiments, the material layer includes silicon nitride, and the second reactant includes a nitrogen precursor.

[0023] According to a second aspect of the disclosure, an apparatus is provided. The apparatus includes a controller including a processor that is programmed to provide a wafer including a patterned structure having a top, a bottom and a sidewall and form a film on the wafer by a cyclical deposition process. The cyclical deposition process includes a cycle of contacting the wafer with a first reactant including a silicon precursor to form an intermediate layer over the patterned structure of the wafer, generating a plasma including H.sup.+ ions to modify the intermediate layer by delivering the H.sup.+ ions anisotropically towards the top and the bottom of the patterned structure, and contacting the wafer with a second reactant to form a material layer. The silicon precursor includes a silicon-nitrogen bond and includes no halogen. The second reactant includes at least one selected from the group consisting of a nitrogen precursor, an oxygen precursor and a carbon precursor.

[0024] Note that this summary section does not specify every embodiment and/or incrementally novel aspect of the present disclosure or claimed invention. Instead, this summary only provides a preliminary discussion of different embodiments and corresponding points of novelty. For additional details and/or possible perspectives of the invention and embodiments, the reader is directed to the Detailed Description section and corresponding figures of the present disclosure as further discussed below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be increased or reduced for clarity of discussion.

[0026] FIG. 1A shows a schematic of a plasma tool in accordance with some embodiments of the present disclosure.

[0027] FIG. 1B shows a vertical cross-sectional schematic of a wafer in accordance with some embodiments of the present disclosure.

[0028] FIG. 2 shows a block diagram of a film deposition process in accordance with some embodiments of the present disclosure.

[0029] FIGS. 3A and 3B respectively show a vertical cross-sectional schematic and view of a wafer

in accordance with some embodiments of the present disclosure.

[0030] FIGS. 3C and 3D respectively show a vertical cross-sectional schematic and view of a wafer in related examples.

[0031] FIG. 4A shows a schematic of an atomic layer deposition (ALD) process in accordance with some embodiments of the present disclosure.

[0032] FIG. 4B shows an energy graph of the ALD process in FIG. 4A in accordance with some embodiments of the present disclosure.

[0033] FIG. 5A shows an ion energy distribution of a plasma in accordance with some embodiments of the present disclosure.

[0034] FIG. 5B shows an ion energy distribution function of a plasma in accordance with some embodiments of the present disclosure.

[0035] FIG. 5C shows an ion angular distribution function of the plasma in FIG. 5B in accordance with some embodiments of the present disclosure.

[0036] FIG. 6 shows a flow chart of a film deposition process in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0037] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, spatially relative terms, such as “top,” “bottom,” “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0038] The order of discussion of the different steps as described herein has been presented for clarity's sake. In general, these steps can be performed in any suitable order. Additionally, although each of the different features, techniques, configurations, etc. herein may be discussed in different places of this disclosure, it is intended that each of the concepts can be executed independently of each other or in combination with each other. Accordingly, the present invention can be embodied and viewed in many different ways.

[0039] In the drawings, like reference numerals designate identical or corresponding parts throughout the several views. Additionally, as used herein, the words “a,” “an” and the like generally carry a meaning of “one or more”, unless stated otherwise.

[0040] Furthermore, the terms, “approximately”, “approximate”, “about” and similar terms generally refer to ranges that include the identified value within a margin of 20%, 10%, or preferably 5%, and any values therebetween.

[0041] As noted in the Background, scaling efforts are running into greater challenges with three-dimensional (3D) semiconductor structure and topography, especially on the scale of sub-30 nm or even single-digit nanometers. Precise control of film thickness is therefore desirable and crucial. Atomic layer deposition (ALD) is capable of producing very thin, conformal films with control of the thickness and composition of the films possible at the atomic level.

[0042] For a 3D topography such as a trench, a hole or a slit, ALD can be used to form one or more films to fill the gap. However, a void or seam is often seen in the ALD film(s) within the filled space. Ideal V-shaped ALD is difficult in terms of effective topological growth inhibition. Particularly for ALD of silicon nitride (SiN), low temperature SiN gap fill often comes with the problem of air voids or seams in the filled space.

[0043] Techniques herein enable void-free or seam-free ALD deposition by combining relatively low ion energy of very high frequency (VHF) plasma and directional bombardment of hydrogen ions. VHF plasma provides substantially lower ion energy than conventional plasma induced at 13 MHz, 27 MHz, 40 MHz and the like. Hydrogen ions produced by VHF plasma tend to have anisotropic velocity. Therefore, when the hydrogen ions are delivered to a top and a bottom of a 3D topography relative to a sidewall, topologically differentiated growth can be achieved, with improved quality at relatively low temperature.

[0044] According to aspects of the disclosure, a SiN film is formed by first using a silicon precursor that contains silicon (Si) and nitrogen (N) and termination of amine base or hydrocarbon base, excluding halogen atoms. Then in a film modification step, plasma is generated above 100 MHz (e.g. VHF, ultra-high frequency (UHF) or microwave frequency) to reduce ion energy induced damage. The generated plasma gas is configured to generate hydrogen ions (e.g. H^{sup.+} ions) for film modification. H^{sup.+} ions can provide monotonic energy and anisotropic velocity relative to wafer horizon. Particularly, since some silicon precursors require ion bombardment energy to enhance the growth, anisotropic H^{sup.+} ions can thus provide the kinetic energy only on the top and bottom surfaces of a 3D topological shape.

[0045] FIG. 1A shows a schematic of a plasma system (referred to as a system **100** hereinafter) in accordance with some embodiments of the present disclosure. As shown, the system **100** includes at least one plasma processing chamber (referred to as a chamber **110** hereinafter). In the chamber **110**, a wafer **130** can be placed on an electrostatic chuck (ESC) **111**. The chamber **110** can be configured to receive or generate a plasma **115** for the wafer **130**. One or more sensors **119** can be used to characterize/monitor the wafer **130** and/or the plasma **115**.

[0046] The chamber **110** can be coupled to a first reactant source **121** by conduits, pipes or the like and receive a first reactant (e.g. a silicon precursor) from the first reactant source **121**. The first reactant source **121** may be coupled to a manifold, valve control system, mass flow control system or the like to release the first reactant in a gaseous form. In some embodiments, the first reactant is liquid or solid under room temperature and standard atmospheric pressure conditions and can be vaporized within a reactant source vacuum vessel, which may be maintained at or above a vaporizing temperature within a precursor source chamber. Accordingly, the vaporized precursor may be transported with a carrier gas (e.g. an inactive or inert gas) and then fed into the chamber **110** through a conduit. In other embodiments, the first reactant may be a vapor under standard conditions. Accordingly, the first reactant may be stored in a gas cylinder and need no carrier gas.

[0047] Similarly, the chamber **110** can be coupled to a second reactant source **125** and receive a second reactant (e.g. a nitrogen precursor) from the second reactant source **125**. Additionally, the chamber **110** can be coupled to a carrier gas source, an inert gas source, a purge gas source and the like (not shown for simplicity purposes). The descriptions have been provided above and will be omitted herein for simplicity purposes.

[0048] In a non-limiting example, the system **100** is configured to execute atomic layer deposition (ALD) of silicon nitride (SiN) on the wafer **130**. Accordingly, the first reactant source **121** and the second reactant source **125** respectively include a silicon precursor and a nitrogen precursor. The nitrogen precursor can include, but are not limited to, N₂, NH₃ and N₂H₄.

[0049] The silicon precursor includes at least one silicon-nitrogen bond and includes no halogen i.e. fluorine (F), chlorine (Cl), bromine (Br), iodine (I), astatine (At) and tennessine (Ts). The silicon precursor may include at least one substitute group selected from the group consisting of a silyl group, an amine group and an alkyl group. For instance, a silyl group can be bonded to a

nitrogen atom. An amine group can be bonded to a silicon atom. An alkyl group can be bonded to a nitrogen atom and/or a silicon atom.

[0050] In one embodiment, the silicon precursor consists of four elements: silicon (Si), nitrogen (N), hydrogen (H) and carbon (C). The silicon precursor may conform to the formula of $\text{Si}(\text{NR}^{\text{sup.1}}\text{R}^{\text{sup.2}})^{\text{sub.n}}(\text{R}^{\text{sup.3}})^{\text{sub.4-n}}$, where n is 1, 2, 3 or 4. $\text{R}^{\text{sup.1}}$, $\text{R}^{\text{sup.2}}$, and $\text{R}^{\text{sup.3}}$ are each independently a hydrogen atom or an alkyl group that is a straight or branched chain. Independently $\text{R}^{\text{sup.1}}$, $\text{R}^{\text{sup.2}}$, and $\text{R}^{\text{sup.3}}$ are preferably a hydrogen atom or an alkyl group having 1-18 carbon atoms, preferably 1-12 carbon atoms, preferably 1-6 carbon atoms, preferably a methyl group, an ethyl group, a propyl group or a butyl group, straight or branched. Examples include, but are not limited to, bis(tertiary-butylamino) silane (BTBAS), bis(diethylamino)-silane (BDEAS) and bis(dimethylamino)dimethylsilane (BDMADMS). In another embodiment, the silicon precursor consists of three elements: Si, N and H. Examples include, but are not limited to, trisilylamine (TSA). Examples of the silicon precursor discussed herein can be used individually or in any combination.

[0051] Further, the chamber **110** can be coupled to a hydrogen ion source **123** and receive a gas from the hydrogen ion source **123**, similar to descriptions above regarding the first reactant source **121**. The gas can be used to generate a plasma containing hydrogen ions in the chamber **110**. The gas can include $\text{H}_{\text{sub.2}}$ and/or a hydrocarbon. In some embodiments, the hydrocarbon may include an alkane that is a straight or branched chain and conforms to the formula of $\text{C}_{\text{sub.n}}\text{H}_{\text{sub.2n+2}}$, where n is an integer of 1 to 18, preferably 1 to 12, preferably 1 to 6, preferably 1, 2, 3 or 4. The hydrocarbon can optionally include at least one carbon-carbon double bond and/or at least one carbon-carbon triple bond. Preferably, the gas includes $\text{H}_{\text{sub.2}}$ and/or $\text{CH}_{\text{sub.4}}$. In some embodiments, the hydrocarbon may include an arene such as benzene or a derivative thereof. The derivative may include one or more substitute groups on a benzene ring and can have 7 to 10 carbon atoms, preferably 7 or 8 carbon atoms. Such substitute groups can have an alkyl group that is a straight or branched chain having 1, 2, 3 or 4 carbon atoms, optionally having at least one carbon-carbon double bond and/or at least one carbon-carbon triple bond. For instance, the arene can include benzene, toluene, xylene, styrene and/or the like. Examples of the gas discussed herein can be used individually or in any combination.

[0052] In other embodiments, the hydrogen ion source **123** can include a remote plasma source that generates a plasma containing hydrogen ions. Accordingly, the chamber **110** is configured to receive the plasma from the hydrogen ion source **123**.

[0053] Additionally, the sensors **119** can include an optical emission spectroscopy (OES) sensor, a voltage peak-to-peak (VPP) sensor, an ion flux sensor, a mass spectrometer, a temperature sensor, a pressure sensor, a reflectometer, an ellipsometer and/or other sensors as known by one skilled in the art. While shown to be installed on a sidewall of the chamber **110** in the example of FIG. 1A, locations of the sensors **119** are not particularly limited. That is, the sensors **119** can each independently be placed inside or outside the chamber **110**, in contact with, in proximity to, distant from or within the wafer **130**, and the like.

[0054] Further, a controller **140** may optionally be included in the example of FIG. 1A. Components of one or more corresponding plasma tools can be connected to and controlled by the controller **140** that may optionally be connected to a corresponding memory storage unit and user interface (all not shown for simplicity purposes). Various plasma-processing operations can be executed via the user interface, and various plasma processing recipes and operations can be stored in a storage unit. Accordingly, a given wafer can be processed within a plasma chamber with various microfabrication techniques.

[0055] The controller **140** may be coupled to various components of the corresponding plasma tool(s) to receive inputs from and provide outputs to the components. For example, the controller **140** can be configured to receive sensor data from the sensors **119** to monitor gas species and/or the wafer **130** in the chamber **110** and determine flow rates of precursors. The controller **140** can also

be configured to adjust knobs and control settings for the corresponding plasma tool(s), or more specifically the chamber **110**, the first reactant source **121**, the hydrogen ion source **123** and the second reactant source **125**. Of course the adjustment(s) can be manually made as well.

[0056] The controller **140** can be implemented in a wide variety of manners. In one example, the controller **140** is a computer. In another example, the controller **140** includes one or more programmable integrated circuits that are programmed to provide the functionality described herein. For example, one or more processors (e.g. microprocessor, microcontroller, central processing unit, etc.), programmable logic devices (e.g. complex programmable logic device (CPLD)), field programmable gate array (FPGA), etc.), and/or other programmable integrated circuits can be programmed with software or other programming instructions to implement the functionality of a proscribed plasma process recipe. It is further noted that the software or other programming instructions can be stored in one or more non-transitory computer-readable mediums (e.g. memory storage devices, FLASH memory, DRAM memory, reprogrammable storage devices, hard drives, floppy disks, DVDs, CD-ROMs, etc.), and the software or other programming instructions when executed by the programmable integrated circuits cause the programmable integrated circuits to perform the processes, functions, and/or capabilities described herein. Other variations could also be implemented.

[0057] Note that the system **100** may include a capacitively-coupled plasma processing apparatus, inductively-coupled plasma processing apparatus, microwave plasma processing apparatus, electron cyclotron resonance (ECR) plasma processing apparatus, or other types of processing systems or combination of systems. Thus, it will be recognized by those skilled in the art that the techniques described herein may be utilized with any of a wide variety of plasma processing systems. The system **100** can be used for a wide variety of operations including, but not limited to, etching, deposition, cleaning, plasma polymerization, plasma-enhanced chemical vapor deposition (PECVD), atomic layer deposition (ALD), atomic layer etch (ALE), and the like. Particularly, ALD is used in this disclosure for illustrative purposes and is not limiting. Structures of plasma tools and ALD tools are both well known to one skilled in the art. It will be recognized that different and/or additional plasma process systems may be implemented while still taking advantage of the techniques described herein.

[0058] FIG. **1B** shows a vertical cross-sectional schematic of the wafer **130** in accordance with some embodiments of the present disclosure. As illustrated, the wafer **130** can include a substrate **131** and a patterned structure **133** formed thereon. The patterned structure **133** can have a top **135**, a bottom **137** and a sidewall **139**. The top **135** and the bottom **137** of the patterned structure **133** are substantially parallel to each other. The patterned structure **133** can include a trench, a hole, a slit, a gap, an opening or any other topography that has a vertical cross-sectional view as shown. For example, the patterned structure **133** can include parallel lines of material (e.g. metal and/or dielectric) with gaps in between. The patterned structure **133** can have a high aspect ratio in that a ratio of a depth D of the patterned structure **133** to a width W of the bottom **137** is in a range of 3 to 20, such as 3, 5, 7, 10, 12, 15, 18, 20 or any value therebetween.

[0059] In one embodiment, the substrate **131** and the patterned structure **133** include different materials. The wafer **130** may further include one or more layers (not shown) formed between the substrate **131** and the patterned structure **133**. In another embodiment, the substrate **131** and the patterned structure **133** include the same material. The substrate **131** and the patterned structure **133** can therefore be a monolithic piece. In a non-limiting example, the substrate **131** includes silicon while the patterned structure **133** includes —NH.sub.2 surface groups on the top **135**, the bottom **137** and the sidewall **139**. The material of the patterned structure **133** is not particularly limited and can include, but is not limited to, silicon, silicon oxide, silicon nitride, silicon oxynitride and silicon carbide.

[0060] FIG. **2** shows a block diagram of a process **200** of atomic layer deposition (ALD) of silicon nitride (SiN) in accordance with some embodiments of the present disclosure. The process **200** can

be executed by the system **100** and the like.

[0061] In block **211**, the aforementioned silicon precursor is introduced into the chamber **110** as the first reactant. Trisilylamine (TSA) will be used herein as one example of the silicon precursor for illustrative purposes. As discussed earlier, the patterned structure **133** of the wafer **130** can include —NH.sub.2 surface groups on the top **135**, the bottom **137** and the sidewall **139**. Therefore, TSA can react with the —NH.sub.2 surface groups to form an intermediate layer, which will be explained further in FIGS. **4A** and **4B**. Subsequently in block **213**, the chamber **110** is purged with a purge gas to remove unreacted TSA.

[0062] In block **221**, a plasma containing hydrogen ions (e.g. H.sup.+ , H.sub.2.sup.+ and H.sub.3.sup.+ ions) is generated to modify the intermediate layer by delivering the hydrogen ions anisotropically towards the top **135** and the bottom **137** of the patterned structure **133**. As shown in FIG. **3A**, a wafer **300A** includes a substrate **331** and a patterned structure **333** having a top **335**, a bottom **337** and a sidewall **339**. The patterned structure **333** corresponds to the patterned structure **133**. As illustrated, H.sup.+ ions **345** can be delivered substantially in the -Z direction, therefore selectively bombarding the top **341** and the bottom **337**, relative to the sidewall **339**. Since the anisotropic ion bombardment of the hydrogen ions can promote film growth, film growth rate is higher at the top **335** and the bottom **337** than at the sidewall **339**. Therefore, more material **340** can be formed at the top **335** and the bottom **337** than at the sidewall **339**.

[0063] Referring back to FIG. **2**, in block **223**, a nitrogen precursor is introduced into the chamber **110** as the second reactant. The nitrogen precursor can react with the intermediate layer to form a material layer of SiN. Note that block **221** and block **223** can be executed simultaneously within block **220**. Alternatively, block **221** can be started or executed prior to block **223**.

[0064] In block **231**, the chamber **110** is purged with a purge gas to remove the (remaining unreacted) nitrogen precursor. A first cycle of ALD is thus accomplished by blocks **211**, **213**, **221**, **223** and **231**. The process **200** may then return to block **211** to repeat the first cycle for at least one more time to form a film. The film may be thicker at the top **135** and the bottom **137** than at the sidewall **139** due to different film growth rates. For example as shown in FIG. **3B**, a film **340'** is thicker at a top **335'** and a bottom **337'** of a patterned structure **333'** than at a sidewall **339'** of the patterned structure **333'**. Note that similar or identical components are labeled with similar or identical numerals unless specified otherwise. For example, the patterned structure **333'** corresponds to the patterned structure **333**.

[0065] Referring back to FIG. **2**, the process **200** can be terminated when a predetermined number of ALD cycles is finished, when a target local film thickness is reached, or when the patterned structure **133** has been filled or overfilled. In some embodiments, a second cycle can be accomplished by blocks **211**, **213**, **223** and **231**. The process **200** may include both the first cycle and the second cycle executed and repeated in any sequence or order.

[0066] While SiN is shown herein for illustrative purposes, it should be understood that techniques herein are also applicable to other films including, but not limited to, silicon oxide, silicon oxynitride and silicon carbide. Accordingly in block **223**, another precursor, such as an oxygen precursor and/or a carbon precursor, may be introduced into the chamber **110**.

[0067] In some embodiments, the controller **140** may optionally be coupled to various components of the process **200** to receive inputs from and provide outputs to the components. For example, the controller **140** can be configured to receive gas flow rate data from blocks **211**, **213**, **221**, **223** and **231** as well as control gas flow rates in blocks **211**, **213**, **221**, **223** and **231**. That is to say, the controller **140** can be configured to implement and monitor blocks **211**, **213**, **221**, **223** and **231**. While not shown, the controller **140** can also be configured to determine when to terminate the process **200** by determining whether the predetermined number of ALD cycles is finished, when the target local film thickness is reached by measurement of the sensors **119**, or when the patterned structure **133** has been filled or overfilled by measurement of the sensors **119**. Of course, one or more functions of the controller **140** can also be manually accomplished.

[0068] FIG. 3C shows a vertical cross-sectional schematic of a wafer **300B** in related examples. A conformal film **360** is grown over a patterned structure **350**. That is to say, the conformal film **360** has a substantially identical film thickness along the patterned structure **350** including a top **355**, a bottom **357** and a sidewall **359**. For example, the conformal film **360** can be a SiN film formed by ALD using precursors of dichlorosilane (DCS) and NH_3 . The ALD chemistry of DCS has a different mechanism in which film growth is driven by radicals **370**. Particularly, DCS is more stable in a chemisorbed state than in a physisorbed state (in contrast to TSA as will be explained in detail in FIGS. 4A and 4B). Therefore as shown in FIG. 3D, a conformal film **360'** has a substantially identical film thickness along a patterned structure **350'** including a top **355'**, a bottom **357'** and a sidewall **359'**.

[0069] FIG. 4A shows a schematic of an absorption process **400A** of SiN during ALD, and FIG. 4B shows an energy graph **400B** of the absorption process in FIG. 4A in accordance with some embodiments of the present disclosure. As shown in state (a), —NH_2 groups are present on a surface, for example on the top **135**, the bottom **137** and the sidewall **139** of the patterned structure **133**. Some —NH_2 groups may be adjacent to —NH groups. State (a) has an energy of 0 eV. [0070] In state (b), a molecule of trisilylamine (TSA) is physically absorbed to the —NH_2 groups. State (b) denoting physisorption has an energy of -0.57 eV. Physisorbed TSA cannot react with —NH_2 groups in a thermal process.

[0071] In state (c), a silicon-nitrogen bond is formed between a TSA molecule and a —NH_2 group. As a result, an H atom of the —NH_2 group may migrate towards a neighboring-NH group. State (c) has an energy of -0.07 eV.

[0072] In state (d), the —NH_2 group loses an H atom to the neighboring-NH group. State (d) has an energy of 0.10 eV.

[0073] In state (e), a gaseous molecule of $\text{HN}(\text{SiH}_3)_2$ is produced and released, leaving a —HN—SiH_3 group on the surface. State (e) denoting chemisorption has an energy of -0.47 eV.

[0074] As discussed earlier in block **211**, TSA can react with —NH_2 surface groups to form an intermediate layer of an ALD cycle. Specifically, the intermediate layer can include states (a), (b), (c), (d) and (e), with state (b) in the most abundance in an equilibrium state due to its lowest energy. Nevertheless, physisorbed TSA in state (b) cannot react with the nitrogen precursor in block **223** whereas chemisorbed TSA in state (e) can.

[0075] State (b) is more stable and favorable than state (e) since state (b) has a lower energy than state (e). Additionally, an activation energy of converting from physisorption (b) to chemisorption (e) is $E_{\text{sub.a}} = 0.10 \text{ eV} - (-0.57 \text{ eV}) = 0.67 \text{ eV}$, which is a relatively big energy barrier and thus further favors state (b). In other words, TSA will be easily forced to go backwards from a chemisorbed state (e) to a physisorbed state (b).

[0076] However, H^+ ions generated in block **221** in FIG. 2 and/or from the hydrogen ion source **123** in FIG. 1A can shift the equilibrium from state (b) to state (e), therefore enhancing the reaction with the nitrogen precursor in block **223** and promoting film growth.

[0077] FIG. 5A shows an ion energy distribution **500A** of a plasma in accordance with some embodiments of the present disclosure. Data were collected by ion mass spectroscopy under conditions of 200 MHz, 600 mTorr, 1000 standard cubic centimeter per minute (sccm) of Ar, 625 sccm of N_2 and 25 sccm of CH_4 . Lines **501**, **503** and **505** respectively represent Ar^+ ions, N_2^+ ions and H^+ ions. As can be seen, H^+ ions have a substantially monotonic energy distribution. In other words, the H^+ ions have a relatively narrow energy distribution compared with Ar^+ ions and N_2^+ ions. An average energy of the H^+ ions is about 18 eV, indicating no or little collision in the sheath.

[0078] FIG. 5B shows ion kinetic simulation results of an ion energy distribution function (IEDF) **500B** of a plasma in accordance with some embodiments of the present disclosure. Lines **511**, **513**, **521**, **523** and **525** respectively represent Ar^+ ions, N_2^+ ions, H^+ ions,

H.sub.2.sup.+ ions and H.sub.3.sup.+ ions. Similarly, the H.sup.+ ions can have a substantially monotonic energy distribution. An average energy of the H.sup.+ ions is about 28 eV. At least 80%, preferably at least 85%, preferably at least 90%, preferably at least 95% of the H.sup.+ ions are within an energy range of 25-31 eV.

[0079] FIG. 5C shows ion kinetic simulation results of an ion angular distribution function (IADF) **500C** of the plasma in FIG. 5B in accordance with some embodiments of the present disclosure. Lines **531**, **533**, **541**, **543** and **545** respectively represent Ar.sup.+ ions, N.sub.2.sup.+ ions, H.sup.+ ions, H.sub.2.sup.+ ions and H.sub.3.sup.+ ions. The H.sup.+ ions have an anisotropic angular distribution. More specifically, the H.sup.+ ions can be substantially unidirectional, with an average angle of about 0°. At least 80%, preferably at least 85%, preferably at least 90%, preferably at least 95% of the H.sup.+ ions are within an angular range of -5° to 5°.

[0080] As can be seen in FIGS. 5A-5C, H.sup.+ ions dominate the hydrogen ions (e.g. H.sup.+, H.sub.2.sup.+ and H.sub.3.sup.+ ions), have a substantially monotonic energy distribution and hardly collide in the sheath. H.sup.+ ions are substantially unidirectional and can thus enable anisotropic bombardment of the top **135** and the bottom **137**.

[0081] Note that the plasma containing the H.sup.+ ions is generated at or above 100 MHz (e.g. very high frequency (VHF), ultra-high frequency (UHF) or microwave frequency) to reduce ion energy induced damage of a surface layer (e.g. **340** in FIG. 3A and state (e) in FIG. 4A). Plasma of a higher frequency tends to result in lower energy as well as better unidirectionality of the H.sup.+ ions. The plasma can have a frequency of 100 MHz to 10 GHz such as 100 MHz, 150 MHz, 200 MHz, 300 MHz, 400 MHz, 500 MHz, 600 MHz, 800 MHz, 1 GHz, 5 GHz, 10 GHz, or any value therebetween. The plasma can be generated at a temperature of -273.15° C. to 500° C. such as -273.15° C., -200° C., -100° C., 0° C., 100° C., 200° C., 300° C., 400° C., 450° C., 500° C., or any value therebetween.

[0082] FIG. 6 shows a flow chart of a process **600** of film deposition in accordance with some embodiments of the present disclosure. At step **S610**, a wafer is provided that includes a patterned structure having a top, a bottom and a sidewall. At step **S620**, the wafer is contacted with a first reactant including a silicon precursor to form an intermediate layer over the patterned structure of the wafer. The silicon precursor includes a silicon-nitrogen bond and includes no halogen. At step **S630**, a plasma including H.sup.+ ions is generated to modify the intermediate layer by delivering the H.sup.+ ions anisotropically towards the top and the bottom of the patterned structure. At step **S640**, the wafer is contacted with a second reactant to form a material layer. The second reactant includes at least one selected from the group consisting of a nitrogen precursor, an oxygen precursor and a carbon precursor. A film can be formed on the wafer by a cyclical deposition process including a cycle of step **S620**, step **S630** and step **S640**.

[0083] In the preceding description, specific details have been set forth, such as a particular geometry of a processing system and descriptions of various components and processes used therein. It should be understood, however, that techniques herein may be practiced in other embodiments that depart from these specific details, and that such details are for purposes of explanation and not limitation. Embodiments disclosed herein have been described with reference to the accompanying drawings. Similarly, for purposes of explanation, specific numbers, materials, and configurations have been set forth in order to provide a thorough understanding. Nevertheless, embodiments may be practiced without such specific details. Components having substantially the same functional constructions are denoted by like reference characters, and thus any redundant descriptions may be omitted.

[0084] Various techniques have been described as multiple discrete operations to assist in understanding the various embodiments. The order of description should not be construed as to imply that these operations are necessarily order dependent. Indeed, these operations need not be performed in the order of presentation. Operations described may be performed in a different order than the described embodiment. Various additional operations may be performed and/or described

operations may be omitted in additional embodiments.

[0085] “Substrate” or “wafer” as used herein generically refers to an object being processed in accordance with the invention. The substrate may include any material portion or structure of a device, particularly a semiconductor or other electronics device, and may, for example, be a base substrate structure, such as a semiconductor wafer, reticle, or a layer on or overlying a base substrate structure such as a thin film. Thus, substrate is not limited to any particular base structure, underlying layer or overlying layer, patterned or un-patterned, but rather, is contemplated to include any such layer or base structure, and any combination of layers and/or base structures. The description may reference particular types of substrates, but this is for illustrative purposes only.

[0086] The substrate can be any suitable substrate, such as a silicon (Si) substrate, a germanium (Ge) substrate, a silicon-germanium (SiGe) substrate, and/or a silicon-on-insulator (SOI) substrate. The substrate may include a semiconductor material, for example, a Group IV semiconductor, a Group III-V compound semiconductor, or a Group II-VI oxide semiconductor. The Group IV semiconductor may include Si, Ge, or SiGe. The substrate may be a bulk wafer or an epitaxial layer.

[0087] Those skilled in the art will also understand that there can be many variations made to the operations of the techniques explained above while still achieving the same objectives of the invention. Such variations are intended to be covered by the scope of this disclosure. As such, the foregoing descriptions of embodiments of the invention are not intended to be limiting. Rather, any limitations to embodiments of the invention are presented in the following claims.

Claims

1. A method of film deposition, the method comprising: providing a wafer including a patterned structure having a top, a bottom and a sidewall; and forming a film on the wafer by a cyclical deposition process comprising a cycle of: contacting the wafer with a first reactant comprising a silicon precursor to form an intermediate layer over the patterned structure of the wafer, wherein the silicon precursor comprises a silicon-nitrogen bond and includes no halogen; generating a plasma comprising H_{sup.+} ions to modify the intermediate layer by delivering the H_{sup.+} ions anisotropically towards the top and the bottom of the patterned structure; and contacting the wafer with a second reactant to form a material layer, the second reactant including at least one selected from the group consisting of a nitrogen precursor, an oxygen precursor and a carbon precursor.
2. The method of claim 1, wherein: the silicon precursor consists of silicon (Si), nitrogen (N), hydrogen (H) and carbon (C) or consists of Si, N and H.
3. The method of claim 2, wherein: the silicon precursor includes at least one of trisilylamine (TSA) or a molecule conforming to a formula of Si(NR_{sup.1}R_{sup.2})_{sub.n}(R_{sup.3})_{sub.4-n}, where n is 1, 2, 3 or 4, and R_{sup.1}, R_{sup.2}, and R_{sup.3} are each independently a hydrogen atom or an alkyl group having 1-18 carbon atoms.
4. The method of claim 3, wherein: the silicon precursor includes at least one selected from the group consisting of TSA, bis(tertiary-butylamino) silane (BTBAS), bis(diethylamino) silane (BDEAS) and bis(dimethylamino)dimethylsilane (BDMADMS).
5. The method of claim 1, further comprising: generating the plasma from a gas including at least one selected from the group consisting of H_{sub.2} and a hydrocarbon.
6. The method of claim 5, wherein: the gas includes at least one selected from the group consisting of H_{sub.2}, benzene, a benzene derivative having 7 to 10 carbon atoms, and an alkane having 1 to 18 carbon atoms.
7. The method of claim 6, wherein: the gas includes H_{sub.2}, CH_{sub.4} or a combination thereof.
8. The method of claim 5, wherein: the plasma further comprises H_{sub.2}_{sup.+} and H_{sub.3}_{sup.+} ions.
9. The method of claim 1, further comprising: generating the plasma at a frequency of 100 MHz to

10 GHz.

10. The method of claim 9, wherein: the H.sup.+ ions are substantially unidirectional so that the H.sup.+ ions are delivered substantially towards the top and the bottom of the patterned structure, relative to the sidewall of the patterned structure.

11. The method of claim 10, wherein: the H.sup.+ ions have a substantially monotonic energy distribution that is configured to promote film deposition at the top and the bottom of the patterned structure.

12. The method of claim 1, wherein: the cyclical deposition process includes repeating the cycle for at least one more time.

13. The method of claim 12, wherein: the film is thicker at the top and the bottom of the patterned structure than at the sidewall of the patterned structure.

14. The method of claim 1, wherein: the plasma is generated before or while the wafer is contacted with the second reactant.

15. The method of claim 1, wherein: the cyclical deposition process is executed at a temperature of no higher than 500° C.

16. The method of claim 1, wherein: the cyclical deposition process comprises atomic layer deposition.

17. The method of claim 1, further comprising: forming —NH.sub.2 surface groups on the patterned structure of the wafer.

18. The method of claim 1, wherein: the material layer comprises one selected from the group consisting of silicon nitride, silicon oxide, silicon oxynitride and silicon carbide.

19. The method of claim 18, wherein: the material layer comprises silicon nitride, and the second reactant comprises a nitrogen precursor.

20. An apparatus, comprising a controller including a processor that is programmed to: provide a wafer including a patterned structure having a top, a bottom and a sidewall; and form a film on the wafer by a cyclical deposition process comprising a cycle of: contacting the wafer with a first reactant comprising a silicon precursor to form an intermediate layer over the patterned structure of the wafer, wherein the silicon precursor comprises a silicon-nitrogen bond and includes no halogen; generating a plasma comprising H.sup.+ ions to modify the intermediate layer by delivering the H.sup.+ ions anisotropically towards the top and the bottom of the patterned structure; and contacting the wafer with a second reactant to form a material layer, the second reactant including at least one selected from the group consisting of a nitrogen precursor, an oxygen precursor and a carbon precursor.
